

Milestone 1 Output

```
df = pd.read_csv(r"/content/us_perm_visas.csv")
print(df)
```

```
...   add_these_pw_job_title_9089  agent_city  agent_firm_name  agent_state \
0                                NaN         NaN             NaN         NaN
1                                NaN         NaN             NaN         NaN
2                                NaN         NaN             NaN         NaN
3                                NaN         NaN             NaN         NaN
4                                NaN         NaN             NaN         NaN
...
2408                               NaN         NaN             NaN         NaN
2409                               NaN         NaN             NaN         NaN
2410                               NaN         NaN             NaN         NaN
2411                               NaN         NaN             NaN         NaN
2412                               NaN         NaN             NaN         NaN

   application_type  case_no  case_number  case_received_date \
0             PERM  A-07322-97014         NaN             NaN
1             PERM  A-07332-99439         NaN             NaN
2             PERM  A-07333-99643         NaN             NaN
3             PERM  A-07339-01930         NaN             NaN
4             PERM  A-07345-03565         NaN             NaN
...
2408             PERM  A-11101-70125         NaN             NaN
2409             PERM  A-11101-70127         NaN             NaN
2410             PERM  A-11101-70143         NaN             NaN
2411             PERM  A-11101-70153         NaN             NaN
2412             PERM  A-11101-70166         NaN             NaN

   case_status  class_of_admission  ...  ri_pvt_employment_firm_to \
0    Certified             J-1  ...                             NaN
1     Denied             B-2  ...                             NaN
2    Certified             H-1B  ...                             NaN
3    Certified             B-2  ...                             NaN
4    Certified             L-1  ...                             NaN
...
2408    Certified             H-1B  ...                             NaN
2409     Denied             B-2  ...                             NaN
2410    Certified             O-1  ...                             NaN
2411    Certified             B-2  ...                             NaN
2412    Certified             H-1B  ...                             NaN

   ri_us_workers_considered  schd_a_shepherd  us_economic_sector \
0                        NaN             NaN                IT
1                        NaN             NaN    Other Economic Sector
2                        NaN             NaN            Aerospace
3                        NaN             NaN    Other Economic Sector
4                        NaN             NaN        Advanced Mfg
...
2408                       NaN             NaN                IT
2409                       NaN             NaN            Construction
2410                       NaN             NaN        Advanced Mfg
2411                       NaN             NaN    Other Economic Sector
2412                       NaN             NaN                NaN

   wage_offer_from_9089  wage_offer_to_9089  wage_offer_unit_of_pay_9089 \
0             75629.00             NaN             yr
1             37024.00             NaN             yr
2             47923.00             NaN             yr
3              10.97             NaN             hr
4            100000.00             NaN             yr
```

```
print("Original Shape:", df.shape)
```

```
Original Shape: (2413, 154)
```

```
# Check missing values
print("\nMissing values:\n", df.isnull().sum())
```

```
...
Missing values:
  add_these_pw_job_title_9089      2413
  agent_city                     2413
  agent_firm_name                 2413
  agent_state                     2413
  application_type                  0
...
  wage_offer_to_9089              1870
  wage_offer_unit_of_pay_9089      1
  wage_offered_from_9089          2413
  wage_offered_to_9089            2413
  wage_offered_unit_of_pay_9089    2413
Length: 154, dtype: int64
```

```
# Save cleaned dataset
# -----
df_encoded.to_csv("visa_milestone1_cleaned.csv", index=False)

print("\nFinal dataset shape:", df_encoded.shape)
print(df_encoded.head())
```

...

Final dataset shape: (2413, 175)

	processing_days	country_ARGENTINA	country_ARMENIA	country_AUSTRALIA	\
0	132	False	True	False	
1	122	False	False	False	
2	44	False	False	False	
3	136	False	False	False	
4	101	False	False	False	

	country_AUSTRIA	country_BANGLADESH	country BARBADOS	country_BELARUS	\
0	False	False	False	False	
1	False	False	False	False	
2	False	False	False	False	
3	False	False	False	False	
4	False	False	False	False	

	country_BELGIUM	country_BELIZE	...	processing_office_TX	\
0	False	False	...	False	
1	False	False	...	False	
2	False	False	...	False	
3	False	False	...	False	
4	False	False	...	False	

	processing_office_UT	processing_office_Unknown	processing_office_VA	\
0	False	False	False	
1	False	False	False	
2	False	False	True	
3	False	False	False	
4	False	False	False	

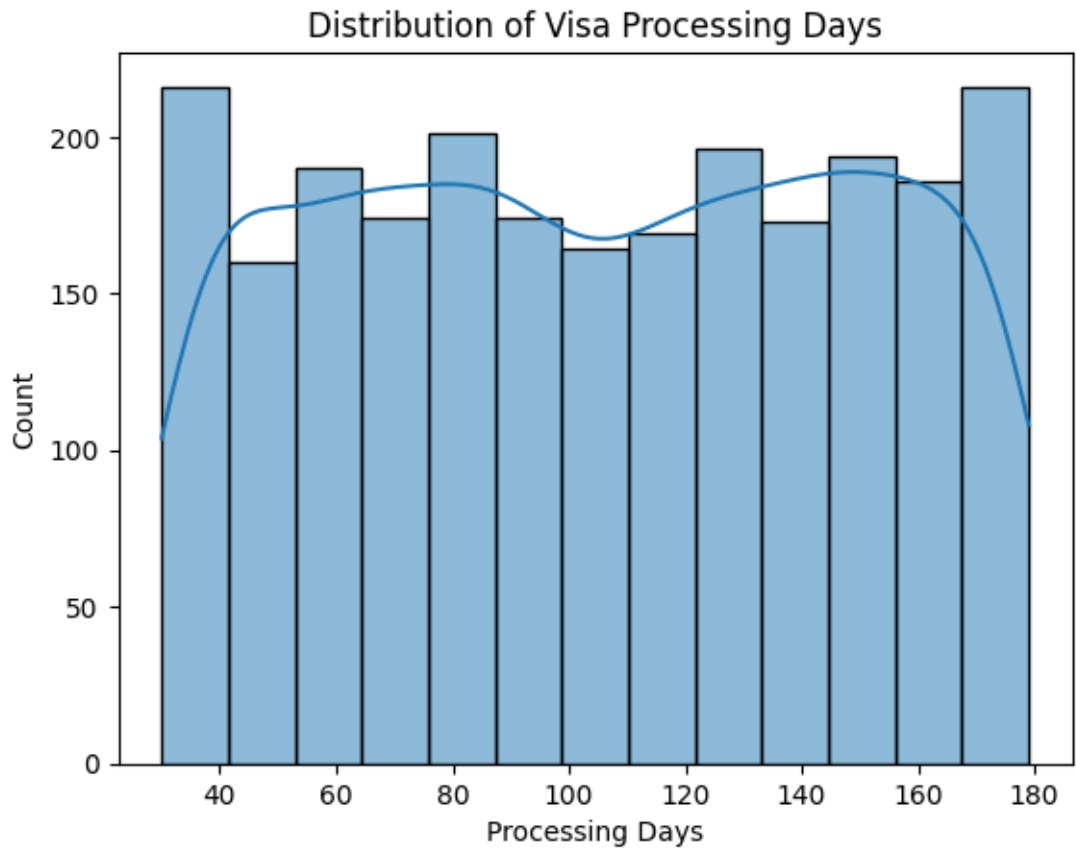
	processing_office_VI	processing_office_VT	processing_office_WA	\
0	False	False	False	
1	False	False	False	
2	False	False	False	
3	False	False	False	
4	False	False	False	

	processing_office_WI	processing_office_WV	processing_office_WY
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False

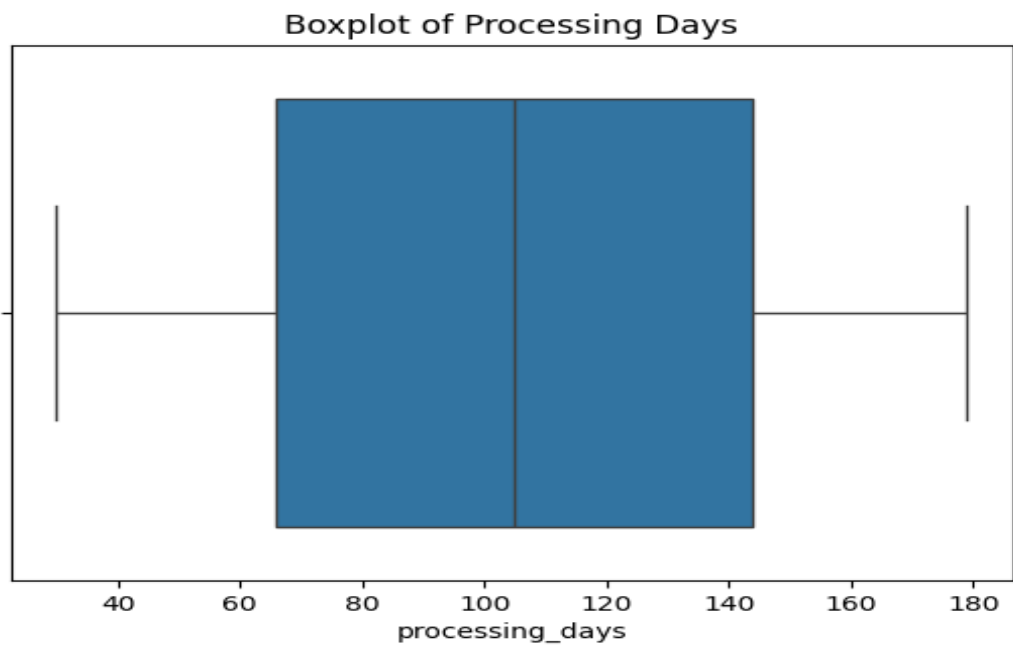
[5 rows x 175 columns]

Milestone 2 Output

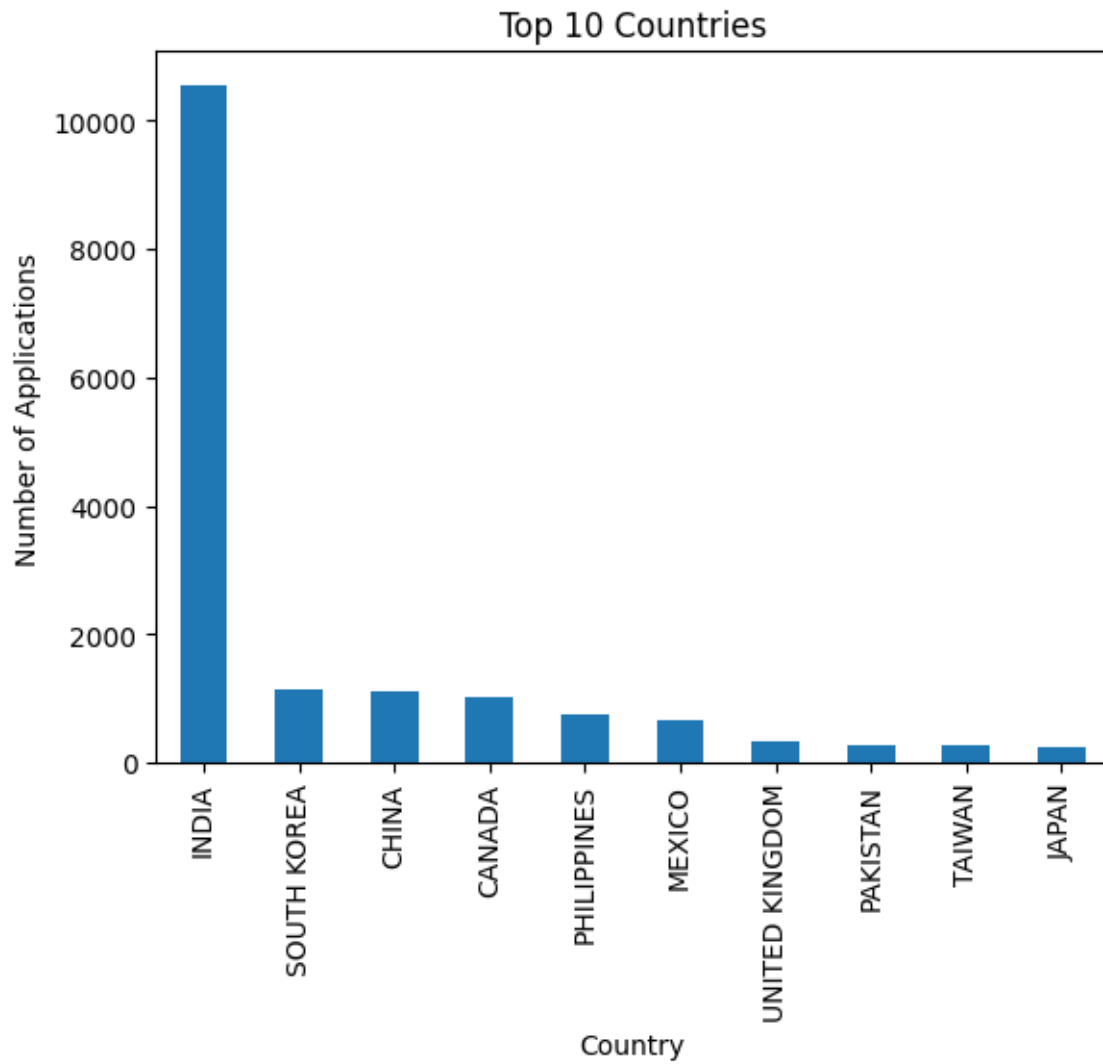
Distribution of Processing Days



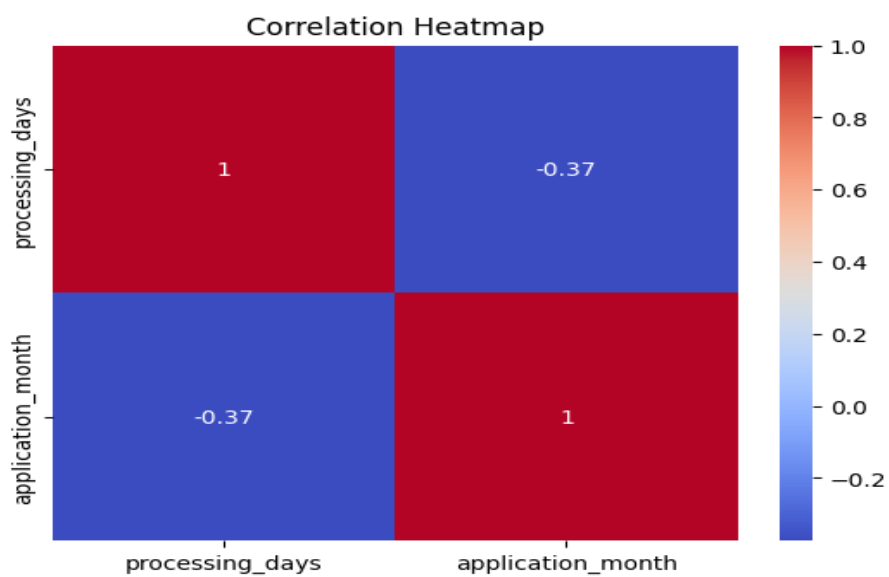
#Boxplot (Outlier Detection)



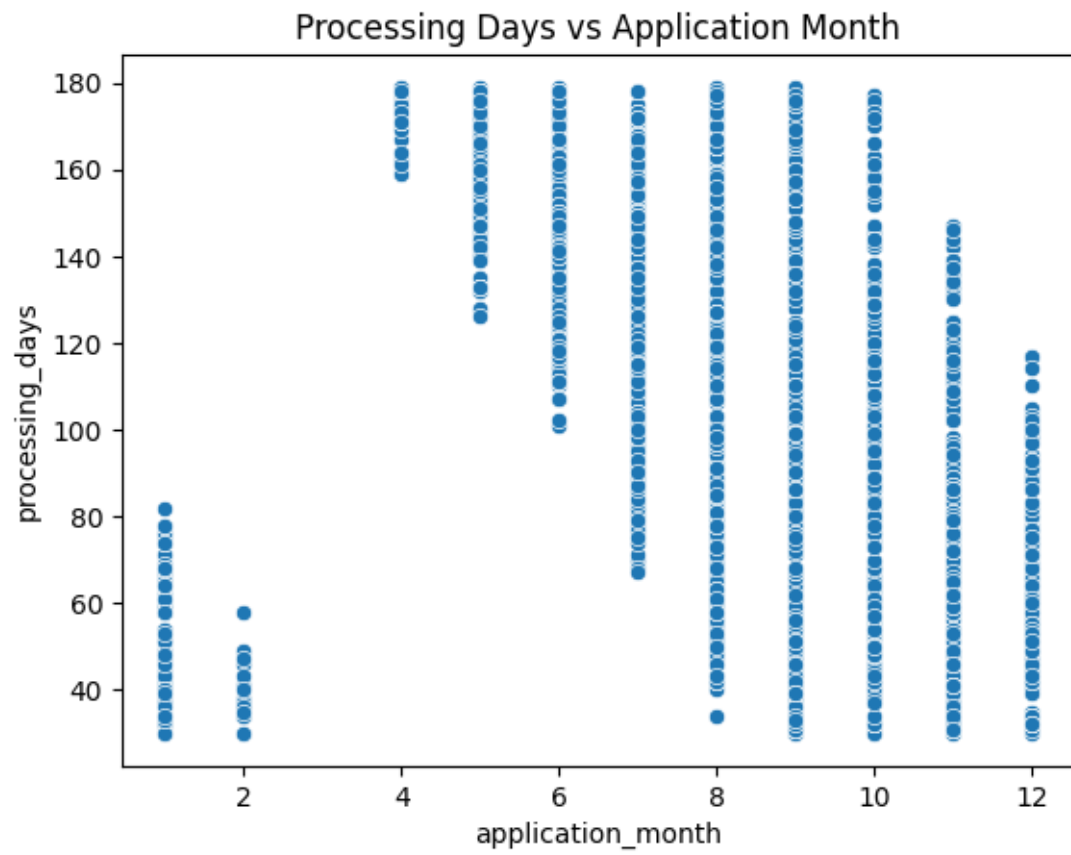
#Top 10 Countries by Applications



correlation heatmap



Scatter Plot



```

#Feature:Application Month
print(df[["application_date", "application_month"]].head())

#Feature:Seasonal Index (Peak vs Off-Peak)
df["season"] = df["application_month"].apply(
    lambda x: "Peak" if x in [1, 2, 12] else "Off-Peak"
)
print(df[["application_month", "season"]].head())

#Feature:Country Specific Average Processing Time
country_avg = df.groupby("country")["processing_days"].mean()
df["country_avg"] = df["country"].map(country_avg)
print("\nCountry-wise Average Processing Time:\n", country_avg.head())

#Feature:Visa Type Average Processing Time
visa_avg = df.groupby("visa_type")["processing_days"].mean()
df["visa_avg"] = df["visa_type"].map(visa_avg)
print("\nVisa-Type Average Processing Time:\n", visa_avg.head())
df.to_csv("visa_milestone2_feature_engineered.csv", index=False)
print(df.head())

```

```

... 4      10 Off-Peak

Country-wise Average Processing Time:
country
Unknown    105.071281
Name: processing_days, dtype: float64

Visa-Type Average Processing Time:
visa_type
A-3      112.000000
A1/A2    107.000000
B-1      106.960000
B-2      103.510417
C-1       72.000000
Name: processing_days, dtype: float64
add_these_pw_job_title_9089 agent_city agent_firm_name agent_state \
0      NaN      NaN      NaN      NaN
1      NaN      NaN      NaN      NaN
2      NaN      NaN      NaN      NaN
3      NaN      NaN      NaN      NaN
4      NaN      NaN      NaN      NaN

application_type case_no case_number case_received_date \
0      PERM  A-07323-97014      NaN      NaN
1      PERM  A-07322-99439      NaN      NaN
2      PERM  A-07333-99643      NaN      NaN
3      PERM  A-07330-01930      NaN      NaN
4      PERM  A-07345-03565      NaN      NaN

case_status visa_type ... wage_offer_unit_of_pay_9089 \
0  Certified    T-1 ...      yr
1  Denied      B-2 ...      yr
2  Certified   H-1B ...      yr
3  Certified    B-2 ...      hr
4  Certified    L-1 ...      yr

wage_offered_from_9089 wage_offered_to_9089 wage_offered_unit_of_pay_9089 \
0      NaN      NaN      NaN
1      NaN      NaN      NaN
2      NaN      NaN      NaN
3      NaN      NaN      NaN
4      NaN      NaN      NaN

processing_days application_date application_month season country_avg \
0      132    2011-09-22      9 Off-Peak    105.071281
1      122    2011-08-21      8 Off-Peak    105.071281
2       44    2011-10-18     10 Off-Peak    105.071281
3      136    2011-07-18      7 Off-Peak    105.071281
4      101    2011-10-17     10 Off-Peak    105.071281

visa_avg
0    108.444444
1    103.510417
2    105.781713
3    103.510417
4    108.013333

[5 rows x 160 columns]

```

```
print(df["country"].value_counts().head(10))
```

```

... country
Unknown    2413
Name: count, dtype: int64

```

```
print(df.columns)
```

```

Index(['add_these_pw_job_title_9089', 'agent_city', 'agent_firm_name',
      'agent_state', 'application_type', 'case_no', 'case_number',
      'case_received_date', 'case_status', 'visa_type',
      ...,
      'wage_offer_unit_of_pay_9089', 'wage_offered_from_9089',
      'wage_offered_to_9089', 'wage_offered_unit_of_pay_9089',
      'processing_days', 'application_date', 'application_month', 'season',
      'country_avg', 'visa_avg'],
      dtype='object', length=160)

```

```

for col in df.columns:
    if "country" in col.lower():
        print(col)

```

```

country
country_of_citizenship
employer_country
foreign_worker_info_birth_country
fw_info_birth_country
country_avg

```